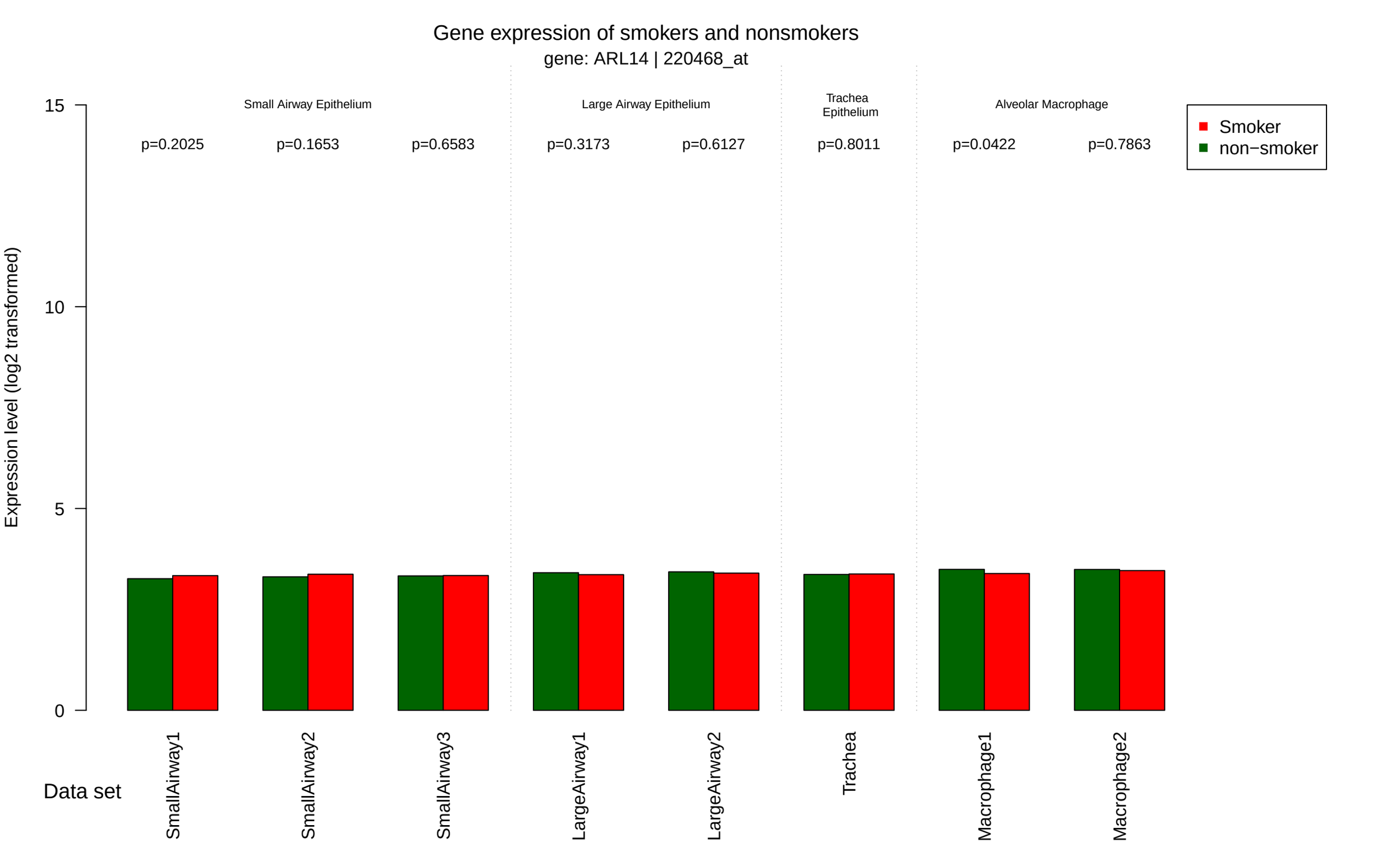
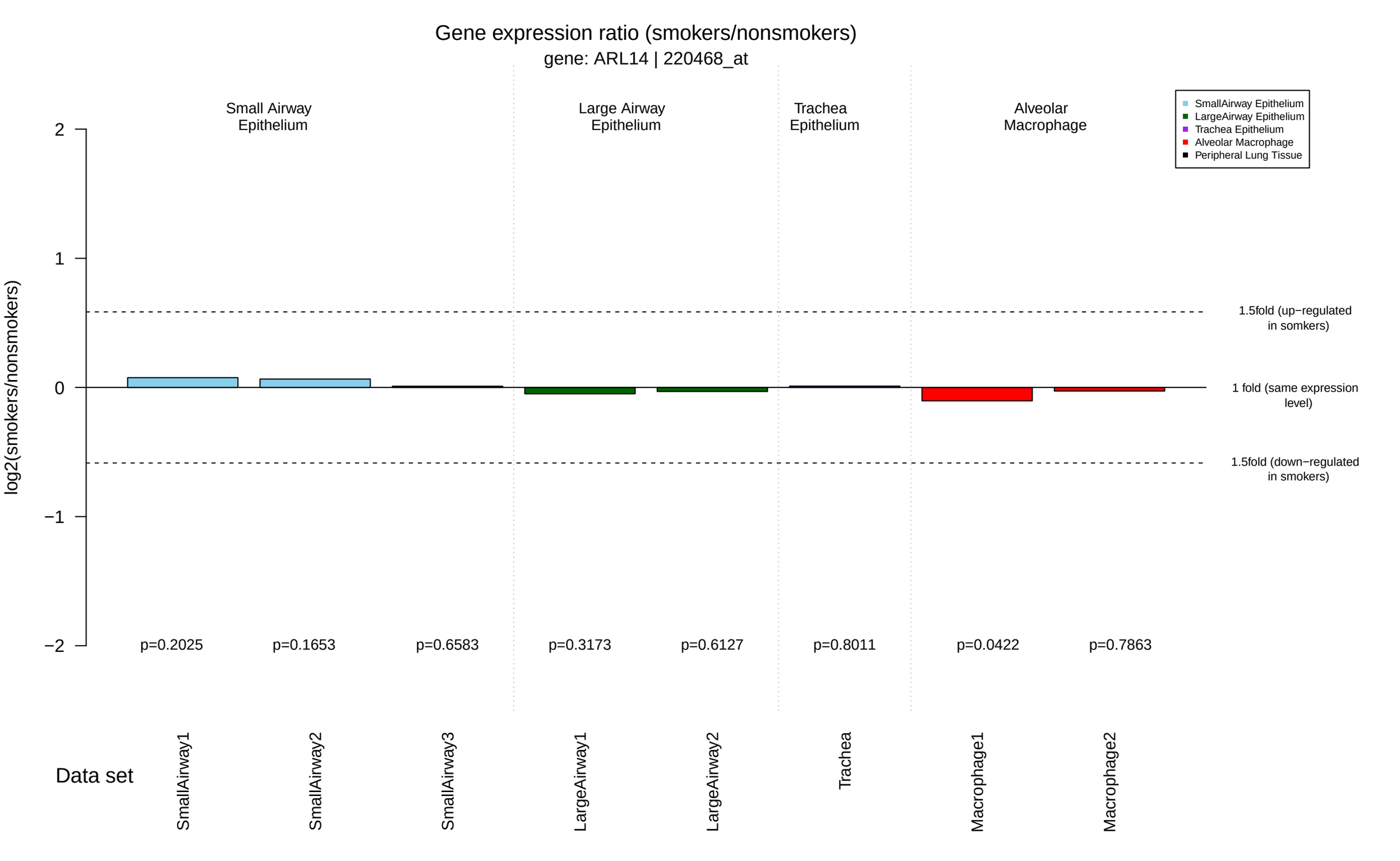
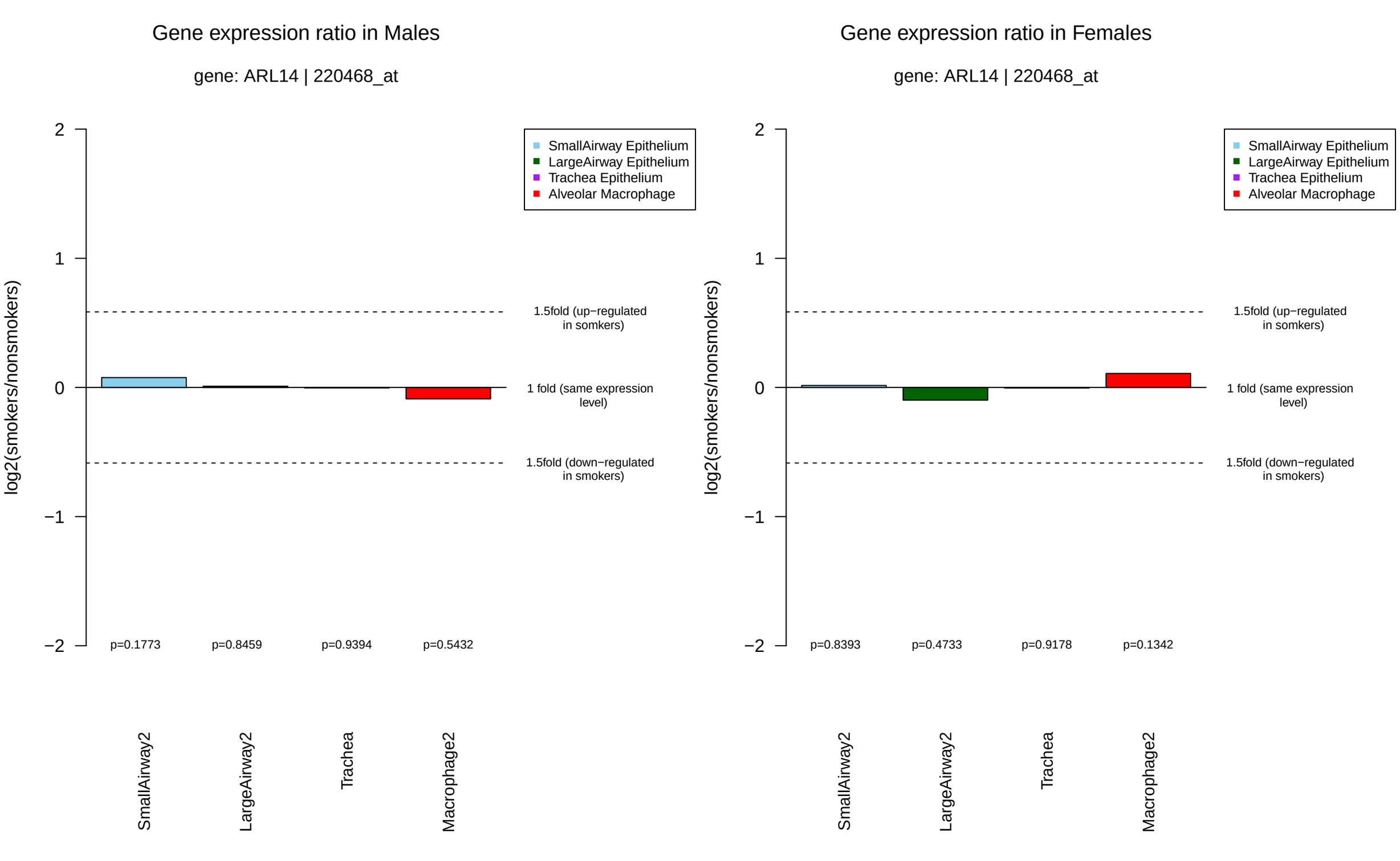
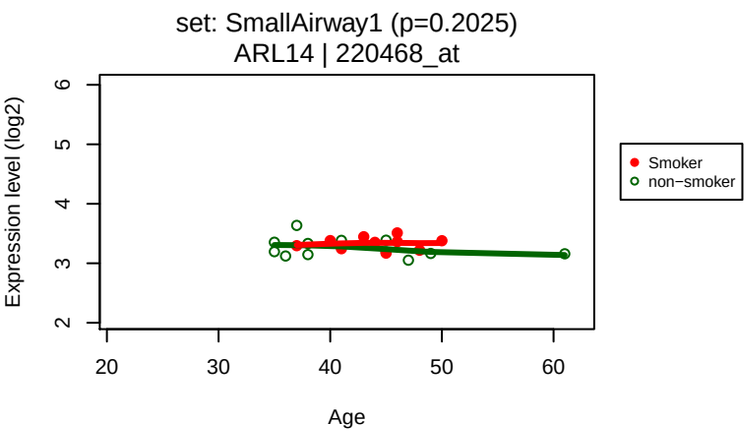
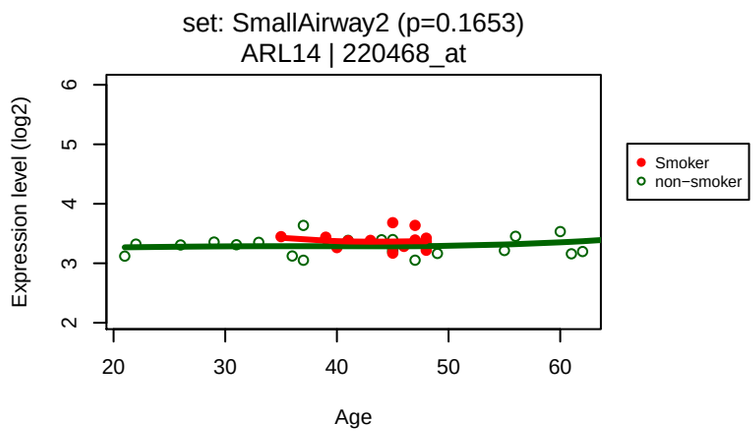
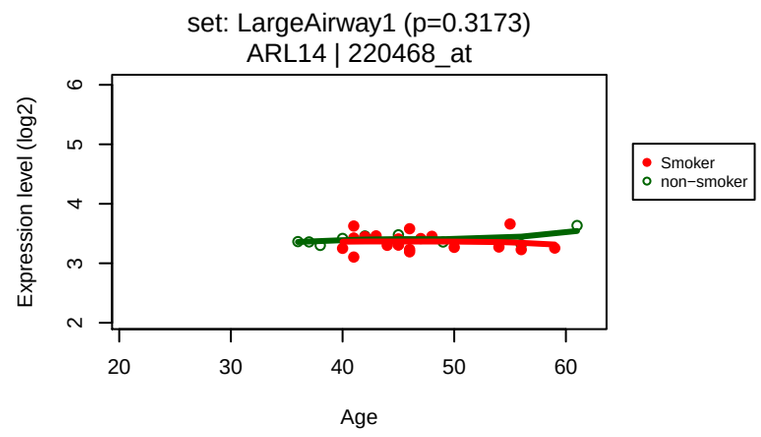

Gene expression of smokers and nonsmokers
gene: ARL14 | 220468_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2025
p=0.1653
p=0.6583
p=0.3173
p=0.6127
p=0.8011
p=0.0422
p=0.7863
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

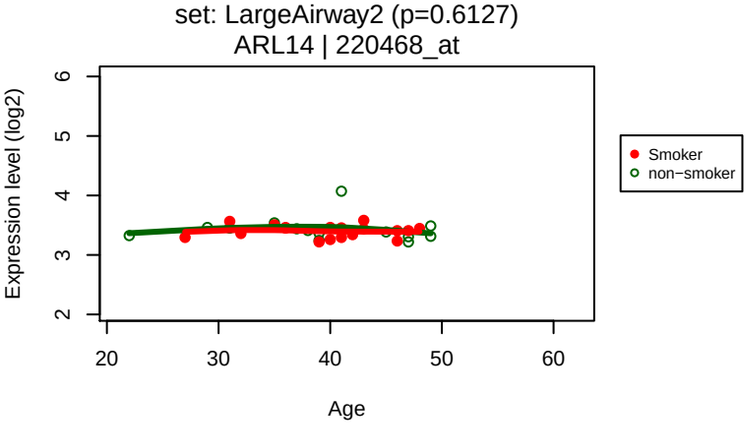

Gene expression ratio (smokers/nonsmokers)
gene: ARL14 | 220468_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2025
p=0.1653
p=0.6583
p=0.3173
p=0.6127
p=0.8011
p=0.0422
p=0.7863
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

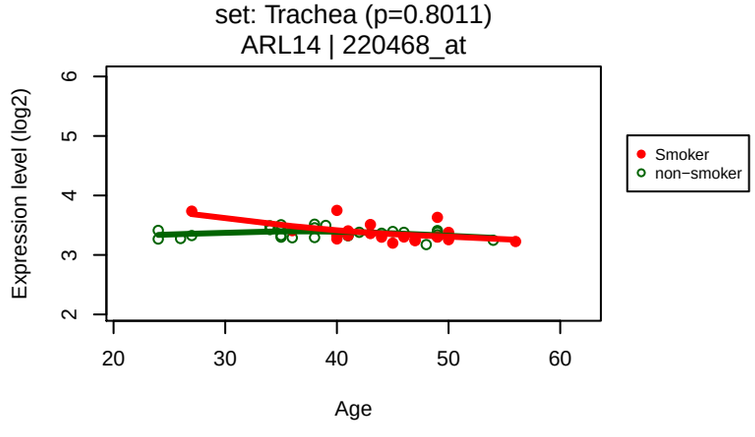

Gene expression ratio in Males
gene: ARL14 | 220468_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1773
p=0.8459
p=0.9394
p=0.5432
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARL14 | 220468_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8393
p=0.4733
p=0.9178
p=0.1342
SmallAirway2
LargeAirway2
Trachea
Macrophage2

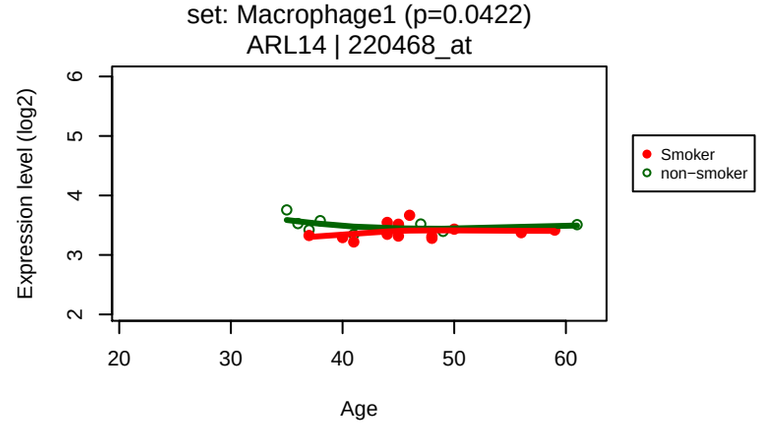

set: SmallAirway1 (p=0.2025)
ARL14 | 220468_at
Expression level (log2)
Age
Smoker
non−smoker

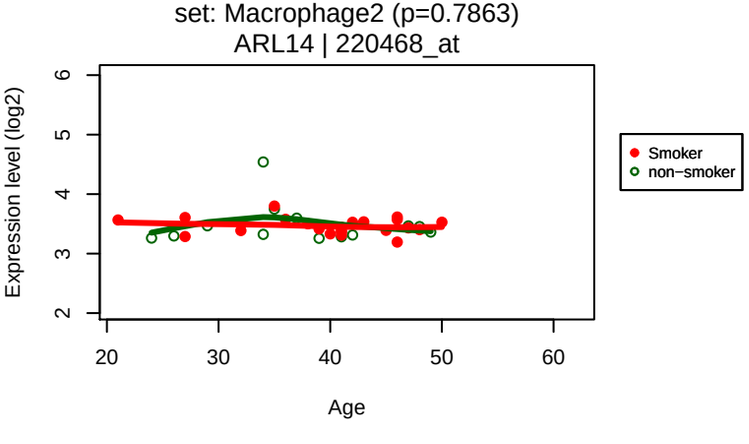

set: SmallAirway2 (p=0.1653)
ARL14 | 220468_at
Expression level (log2)
Age
Smoker
non−smoker

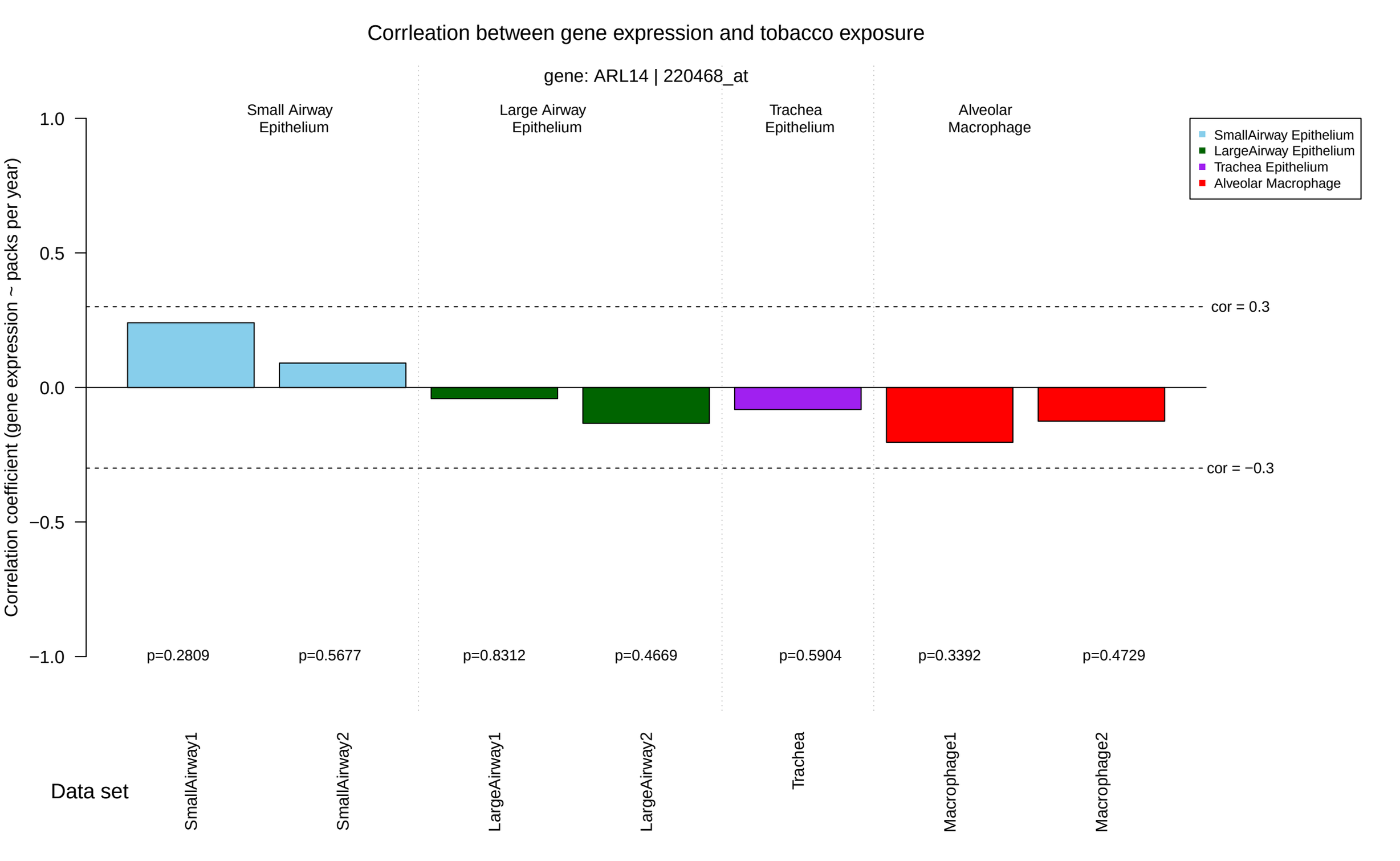

set: LargeAirway1 (p=0.3173)
ARL14 | 220468_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6127)
ARL14 | 220468_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8011)
ARL14 | 220468_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0422)
ARL14 | 220468_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7863)
ARL14 | 220468_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARL14 | 220468_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2809
p=0.5677
p=0.8312
p=0.4669
p=0.5904
p=0.3392
p=0.4729
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2